

EGGSHELL CALCIUM IN THE PREVENTION AND TREATMENT OF OSTEOPOROSIS

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Summary: *In this paper the most significant biological and clinical aspects of a biopreparation made of chicken eggshells are reviewed. Eggshell powder is a natural source of calcium and other elements (e.g. strontium and fluorine) which may have a positive effect on bone metabolism. Experimental and clinical studies performed to date have shown a number of positive properties of eggshell powder, such as antirachitic effects in rats and humans. A positive effect was observed on bone density in animal models of postmenopausal osteoporosis in ovariectomized female rats. In vitro eggshell powder stimulates chondrocyte differentiation and cartilage growth. Clinical studies in postmenopausal women and women with senile osteoporosis showed that eggshell powder reduces pain and osteoresorption and increases mobility and bone density or arrests its loss. The bioavailability of calcium from this source, as tested in piglets, was similar or better than that of food grade purified calcium carbonate. Clinical and experimental studies showed that eggshell powder has positive effects on bone and cartilage and that it is suitable in the prevention and treatment of osteoporosis.*

Historical background

The purpose of the chicken eggshell is to protect the enclosed chicken embryo. It has been used by humans as a source of calcium in powder form added to various mixtures (mixed with alcohol or lemon juice). Local healers used these blends mainly as sup-

portive medications in the treatment of tuberculosis to increase calcification. In the 1940s, Hungarian physician Krompecher and his colleagues developed the basics of therapy using eggshells. Dr. Krompecher was employed by the Biological Research Institute in Tihanyi and started his general practice in the region of the Balaton Lake in 1945–1947. In this practice he came across serious cases of rachitis and, from the means at his disposal, he chose to use eggshells to treat this disease. In extensive metabolic studies and experimental investigations he proved the value of the use of crushed or ground eggshells in the treatment of rachitis. He would administer children with

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rachitis a daily dose of up to 6 g of eggshells. The shells were heat-treated in a simple oven. Observed during the therapy were regression of typical skeletal deformities and improvement of blood tests (increased erythrocyte counts and normalization of erythrocyte sedimentation rate). Treated children gained weight and their physical condition improved. The introduction of eggshells to pregnant women in families with a high occurrence of rachitis reduced the frequencies of rachitic newborns and substantially lowered the rates of congenital hip anomalies. Dr. Krompecher summarized his experience in the books titled "Die Knochenbildung" (1937) (1) and in "Die Grundlage der Eierschalentherapie," published in 1958 (2). Prof. Lichtenstein (1948) (3) notes that eggshell powder is a large store of calcium, which gets wasted instead of being used.

The era of Slovak eggshell powder started in the 1980s when Dr. Karol Michálek discovered a new method of processing eggshells, based on washing, drying, grinding and thermal treatment. In the 1990s, the method was patented in the former Czechoslovakia, various European countries and subsequently in the United States. Chicken eggshell powder was registered as a medicinal drug (Biomin H[®], Biomin a.s., Cífer, Slovak Republic) in 1992. Biomin H[®] is a natural source of calcium. In addition to calcium and phosphorus it also contains trace amounts of other elements and small amounts of organic substances that supplement or potentate the effect of calcium. One gram of the powder contains 370 mg calcium, 0.6 mg phosphorus, 5 mg of magnesium and other elements (Table I).

Composition of chicken eggshell powder

The contents of calcium and other elements in Biomin H[®] oscillate around the values listed in Table I and depend mainly on the season when the eggs

Table I Composition of Biomin H[®]*

Element	Units	Quantity
Nitrogen	mg/g	3.96
Ammonia	mg/g	0.38
Calcium	mg/g	370
Magnesium	mg/g	5.0
Phosphorus	mg/g	0.60
Strontium	µg/g	372
Zinc	µg/g	5.1
Manganese	µg/g	0.2
Iron	µg/g	22.4
Copper	µg/g	5.7
Lithium	µg/g	0.3
Boron	µg/g	< 0.5
Chromium	µg/g	0.12
Fluorine	µg/g	3.75
Selenium	ng/g	23.5
Vanadium	µg/g	< 0.5
Lead	µg/g	< 0.5
Aluminum	µg/g	1.2
Cadmium	ng/g	< 50
Mercury	µg/g	< 0.2

*The mean of six analyses.

were laid (winter, spring, summer or fall). However, seasonal variations do not significantly affect the quality of the prepared powder. The amounts of boron, vanadium, lead, cadmium and mercury are below detection limits. Schaafsma et al. (4) compared the effect of poultry-raising on the contents of eggshell powder prepared in the Netherlands from three different sources of eggs (Table II). Eggshells produced by hens fed on a green grass and by grain and legumes (ES2) contained more strontium, lead and cadmium than shells produced by hens fed on a standard feeding mix. The authors regard this as the result of environmental contamination. Compared with the Dutch eggshells, Slovak eggshell powder is among the sources of calcium with the least heavy metal content, especially cadmium (Table I).

Among the microelements, only strontium is present in significant amounts (372 ± 161 µg/g). Sup-

Table II Influence of chicken feed on the eggshell composition*

Component		ES1	ES2	ES3
Nitrogen	mg/g	4.01	3.90	4.02
Ammonia	mg/g	0.65	0.66	0.68
Calcium	mg/g	385	393	387
Magnesium	mg/g	3.6	3.6	3.5
Phosphorus	mg/g	1.50	0.20	1.90
Strontium	µg/g	380	880	320
Zinc	µg/g	5.0	5.0	4.0
Iron	µg/g	23.0	23.0	22.5
Copper	µg/g	10.0	13.0	7.5
Boron	µg/g	0.5	0.5	0.8
Chromium	µg/g	0.20	0.40	0.25
Fluorine	µg/g	6.0	5.5	3.0
Selenium	ng/g	45.0	50.0	53.0
Vanadium	µg/g	< 0.5	< 0.5	< 0.5
Lead	µg/g	1.10	1.30	0.53
Aluminum	µg/g	2.0	2.0	2.0
Cadmium	ng/g	18.0	24.0	3.0
Mercury	µg/g	< 0.2	< 0.2	< 0.2

ES1 = Dutch "four cereal" egg; ES2 = Dutch "grass" egg; ES3 = Dutch battery egg. *Table adapted from Schaafsma (4). The eggshells analyzed were from Dutch poultry-raising.

plementation of 6 g of eggshell powder per day covers 50% of the daily strontium need. Strontium is a chemical analog of calcium and can be built into bone mass. It has a significant anabolic and antiresorption effect on bone cells (5). Several strontium compounds (e.g. strontium ranelate) are now being introduced to the market as potential drugs for osteoporosis treatment. Several other microelements should be mentioned, such as magnesium, zinc, copper, manganese and fluorine, which also affect the quality of bone mass (6, 7). Magnesium has several effects on bone, influencing bone remodeling, strength and preservation (8). Magnesium is not an integral component of hydroxyapatite crystals but rather it lies on their surface and inhibits their formation. While limiting the size of these crystals, it increases bone strength. Magnesium deficiency results in changes in calcium metabolism, often leading to hypocal-

cemia, vitamin D abnormality and neuromuscular hyperexcitation (9).

The frequent aversion to natural sources of calcium comes from fear of impurities, mainly heavy metals. In some cases (e.g. taking dolomite as a source of calcium), this fear is justified, as the presence of larger quantities of heavy metals has been found in dolomite. However, this does not apply to Biomin H[®]. Dutch scientists presented an interesting finding (4) showing that the concentrations of potential toxic elements such as lead, mercury and cadmium in eggshell powder from Slovakia and Japan were lower than in preparations made of pure calcium carbonate or oyster shells (Table III). The contents of heavy metals in eggshells are far below the acceptable daily intake level, even at the recommended daily calcium dose of 1,200 mg. This is probably due to the uptake of heavy metals in the liver, kidneys and other organs of the laying hens.

Table III Mineral compositions of Slovakian eggshell powder and three commercially available calcium preparations

Component		Slovakian eggshell	Calcipural ¹	Oyster shell powder ²	Japanese eggshell	RDA, DRI* or ADI**
Calcium	mg/g	370-415	397	295	393	1,200 mg*
Magnesium	mg/g	3.5-5.5	< 0.03	78.8	28	320-420 mg*
Phosphorus	mg/g	0.6-1.4	nd	0.15	0.1	700 mg*
Strontium	mg/g	0.050-0.693	0.230	0.910	0.410	1-3 mg
Zinc	mg/g	0.004-0.006	< 0.001	0.005	0.004	12-15 mg
Iron	mg/g	0.020-0.025	0.036	0.42	0.027	10-15 mg
Copper	mg/g	0.005-0.010	0.008	0.007	0.009	1.5-3.0 mg
Boron	µg/g	< 0.5	0.5	9	nd	500-1,000 µg
Chromium	µg/g	0.03-0.20	1.1	1.0	nd	50-200 µg
Fluorine	mg/g	0.002-0.006	0.025	0.070	nd	3.1-3.8 mg*
Selenium	µg/g	0.014-0.034	0.008	0.145	nd	55-70 µg
Vanadium	µg/g	< 0.5	0.5	1	nd	< 10 µg
Lead	µg/g	< 0.5	0.99	0.72	< 0.5	430 µg**
Aluminum	mg/g	< 0.005	0.007	4.4	< 0.005	70 mg**
Cadmium	µg/g	< 0.050	0.657	0.062	< 0.050	60 µg**
Mercury	µg/g	< 0.2	< 0.2	< 0.2	< 0.2	29 µg**

¹Food grade, purified calcium carbonate (Scora S.A., Caffiers, France); ²oyster calcium (Vitasan, OTC Pharma BV, Gorinchem, The Netherlands); Japanese eggshell powder (Furikake Tamaga Calcium, Toya Keiran, Japan). *DRI = dietary reference intake for 50+ years (Standing Committee on the Scientific Evaluation of Dietary Reference Intakes 1997) (48). **ADI = acceptable daily intake for an adult weighing 70 kg (Dokkum van 1985) (49). RDA = recommended dietary allowances for 50+ years (National Research Council 1989) (50).

Table adapted from Schaafsma (4). Nd = not determined.

Calcium in the prevention and treatment of osteoporosis

A calcium-rich diet remains the most suitable source of calcium. This includes mainly milk, dairy products, yogurts, hard cheeses and others because bones also need other nutritional compounds and a balanced supply of minerals and vitamins (10). Some people avoid milk because of its high fat content, others because of lactose intolerance or allergic reactions to dairy products. Long-term studies in several European countries and in the USA showed that a standard diet contains far less than the necessary amount of calcium, which increases the risk of osteoporosis and bone fractures. Consequently, the World Health Organization suggested calcium supple-

ments should be used in the form of dietary calcium additives. Although the recommended doses can vary based on the results of different expert teams, they do not significantly differ from those shown in Table IV. Calcium intake varies with age. Sufficient calcium intake in childhood and puberty plays a significant role in reaching a genetically determined optimal bone mass in young age which will later significantly influence the risk of osteoporosis. Permanent and sufficient intake of calcium increases the accumulation of bone mass in growing children and adolescents (11-13). It is important to maintain bone mass at adult age and minimize its loss at older age.

Since calcium represents a basic part of bone mineral, its administration is an absolute and necessary prerequisite for any treatment of osteoporosis.

Table IV Recommended dietary allowances of calcium during life span

Category		Recommended dietary allowances (mg/day)
Children	0-6 months	400
	6 months - 1 year	600
	1-5 years	800
	6-10 years	800-1,200
Adolescents	11-24 years	1,200-1,500
Males	25-65 years	1,000
	> 65 years	1,500
Females	pregnant and lactating	1,200-1,500
	25 years - menopausal	1,000
	Postmenopausal:	
	with HRT	1,000-1,200
	without HRT	1,500

Calcium monotherapy is generally ineffective in the treatment of postmenopausal high-turnover osteoporosis (14), but even in this case it is a suitable supplement to hormone replacement therapy (15). In peri- and postmenopausal women, supplementation of calcium usually leads to only a small gain of bone density in some sites of the bone skeleton or to a delay in the rate of bone loss (16, 17). Studies of the efficacy of calcium supplementation on the occurrence of fractures in older women have repeatedly proven its positive role in the prevention of vertebral compression (18, 19). Other results confirmed the effect of calcium supplementation on the prevention of proximal femoral fractures. In combination with vitamin D, it is the treatment of choice for senile osteoporosis, particularly in institutionalized patients (20-22).

Calcium absorption from calcium carbonate and from eggshell powder

Calcium is present in eggshells in the form of calcium carbonate. Due to its water insolubility, calcium

carbonate has long been underestimated as a good calcium source and more soluble organic salts were given priority, such as calcium citrate or calcium ascorbate. However, it was subsequently shown that the absorption of calcium from its various salts does not differ significantly. Although calcium carbonate is water insoluble, it is easily ionized at the low pH present in the stomach. Deroisy *et al.* (23) compared the bioavailability of calcium from various salts such as calcium carbonate, calcium gluconolactate, calcium citrate, calcium pidolate and ossein-hydroxyapatite complex in men. The availability of calcium was measured indirectly by studying the increase of serum calcium and decrease of parathormone concentration in sera after a single intake of 500 mg of calcium in the form of the above-mentioned salts. All tested substances increased serum calcium concentration and decreased parathormone concentration. The level of parathormone was most significantly decreased by calcium carbonate, which was also the best source for calcium absorption, followed by calcium citrate and the other tested salts.

Another advantage of calcium carbonate is the favorable ratio of calcium to carbonate residue, which means that a patient having to use 1 g of Ca, for example, needs to take 2.5 g of salt or eventually 2.7 g of eggshell powder; while in the form of calcium lactate, for example, the same patient needs to take 7.7 g of salt. Schaafsma and Beelen (24) studied the bioavailability of calcium from calcium carbonate and eggshell powder in piglets. The digestibility of calcium in piglets is a good proximal method for studying calcium absorption in humans, because intestinal absorption in these animals is the closest to that of humans. Calcium absorption was studied in two diet mixes, in combination with casein or proteins isolated from soya. In both cases calcium absorption was greater from the eggshell powder. Soya contains isoflavones, which have weak estrogenic effects, and thus it has a preventive effect on osteoporosis (25).

On the other hand, soya proteins inhibit calcium absorption. Another important finding is that these proteins inhibit the absorption of calcium from eggshells at a lower rate than calcium from pure calcium carbonate.

Eggshell powder: Growth factors and other biologically active substances

Small amounts of insulin-like growth factor-1 (12 ng/g) and transforming growth factor- β 1 (0.75-7.23 ng/g) were found in eggshell extracts (4). The egg membrane is a richer source of growth factors and its removal during eggshell processing should therefore be reconsidered. Small amounts of calcitonin (10-25 ng/g) and progesterone (0.30-0.33 ng/g) were found in eggshells by Schaafsma *et al.* (4). Eggshell powder also has moderate immunostimulatory properties *in vitro* – it stimulates T-lymphocytes and increases natural killer-cell activity (26).

An enzyme histochemistry study by Pavlíková *et al.* (27) on the effect of eggshell powder on *in vitro* chicken embryonic cartilage growth proved that this substance significantly stimulated cartilage growth. The activity of alkaline phosphatase was studied. This enzyme is present in the cell membrane of chondrocytes and is considered to be a marker of differentiation of these cells. Pelvic cartilage and cartilage of the femur and tibia were incubated *in vitro* for 24 h in the presence of chicken eggshell powder. High alkaline phosphatase activity was present in the perichondrium and around hypertrophic chondrocytes. A significantly thicker layer of hypertrophic chondrocytes showing alkaline phosphatase positivity was observed. Eggshell powder stabilizes the cell membrane of differentiating rounded and hypertrophic chondrocytes and osteoblasts and activates alkaline phosphatase on their cell membranes. It thus enables their differentiation and cartilage and bone growth in *in vitro*

conditions. The presence of eggshell powder in incubation medium of pelvic cartilage of 9-day-old chicken embryo significantly increased the weight of cartilage after 24- and 48-h *in vitro* incubation.

Experimental studies in animal models

Studies in ovariectomized female rats showed that eggshell powder is more than just calcium carbonate. Like postmenopausal women, ovariectomized female rats lost a significant portion of their bone mass after 9 weeks of the study. Since the rats' metabolism is significantly faster than that of humans, bone mass loss was also faster. For this reason, this model was used to study postmenopausal osteoporosis. While animals treated by chemically pure calcium carbonate showed only a small increase of bone mineral density (BMD) and whole body bone mineral content compared with untreated animals, animals treated with chicken eggshell powder had significantly higher density than the untreated ovariectomized female rats (28). Urine pyridinoline and deoxypyridinoline concentrations significantly decreased (29). In another study, the effect of the combination therapy with sodium alendronate plus eggshell powder (30) was studied. Eggshell powder again proved to be very useful. The effect of treatment with a half dose of alendronate in combination with eggshell powder was similar to that of sodium alendronate given alone at a dose required to prevent bone density loss. Addition of vitamin D2 to eggshell powder had no positive effect on BMD in rats (31, 32). However, Hirasawa *et al.* (33) showed that the addition of vitamin D3 increases the beneficial effect of eggshell powder in ovariectomized animals. Interesting results were obtained with a combination of eggshell powder and a small amount of type I collagen, which moderately potentiates the effect of the former (34).

Preclinical and clinical studies in humans

The first preclinical study of eggshell powder (Biomim H[®]), currently regarded as historical, by Makai and Chudáček (35) was published in 1991. A group of patients with osteoporosis, consisting of 52 women and 32 men, was treated with a daily dose of 6 g of eggshell powder for 2-6 months. In most of the patients, reduced pain, improved mobility and increased bone density (as measured by Kocián index) were observed. Pain reduction was also observed in patients with osteoarthritis and slight or no osteoporosis. After these very promising results, physicians and patients have been slowly becoming aware of the therapeutic effects of eggshell powder. The clinical study by Palmaj (36), which also looked into the issue of chicken eggshell powder tolerance, was performed in orthopedic patients with a variety of diagnoses. The author reported very good eggshell powder tolerance, except in patients with hypoacidity. The product had beneficial effects on remineralization of osteoporotic bones, fracture healing and

pseudoarthrosis. A significant analgesic effect of the product was also noted. A comparative study with calcium effervescence (Pharmacia) of Stejskal and Bártek (37) in patients with osteoporosis clearly supported the beneficial effect of eggshell powder. In the study of Masaryk *et al.* (38), 30 women with vertebral osteoporosis and at least one vertebral fracture received 3 g (*i.e.* 1,100 mg of Ca) of chicken eggshell powder once daily (Fig. 1). BMD of L2-L4, the femoral neck, trochanter, and Ward's triangle before and after treatment was measured using a NORLAND XR-26 DEXA densitometer. The following biochemical parameters were studied: osteocalcin, serum calcium and phosphorus, 24-h calciuria and urine pyridinoline and deoxypyridinoline. After 1 year of treatment, Ward's triangle density was increased (+2.77%), with the density values decreasing for the remaining areas studied. Patients aged 70 years or more showed better results, exhibiting density increases in Ward's triangle (+4.51%), the L2-L4 region (+3.26%) and the femoral neck (+1.38%). During treatment, serum calcium level showed a tendency to increase, while

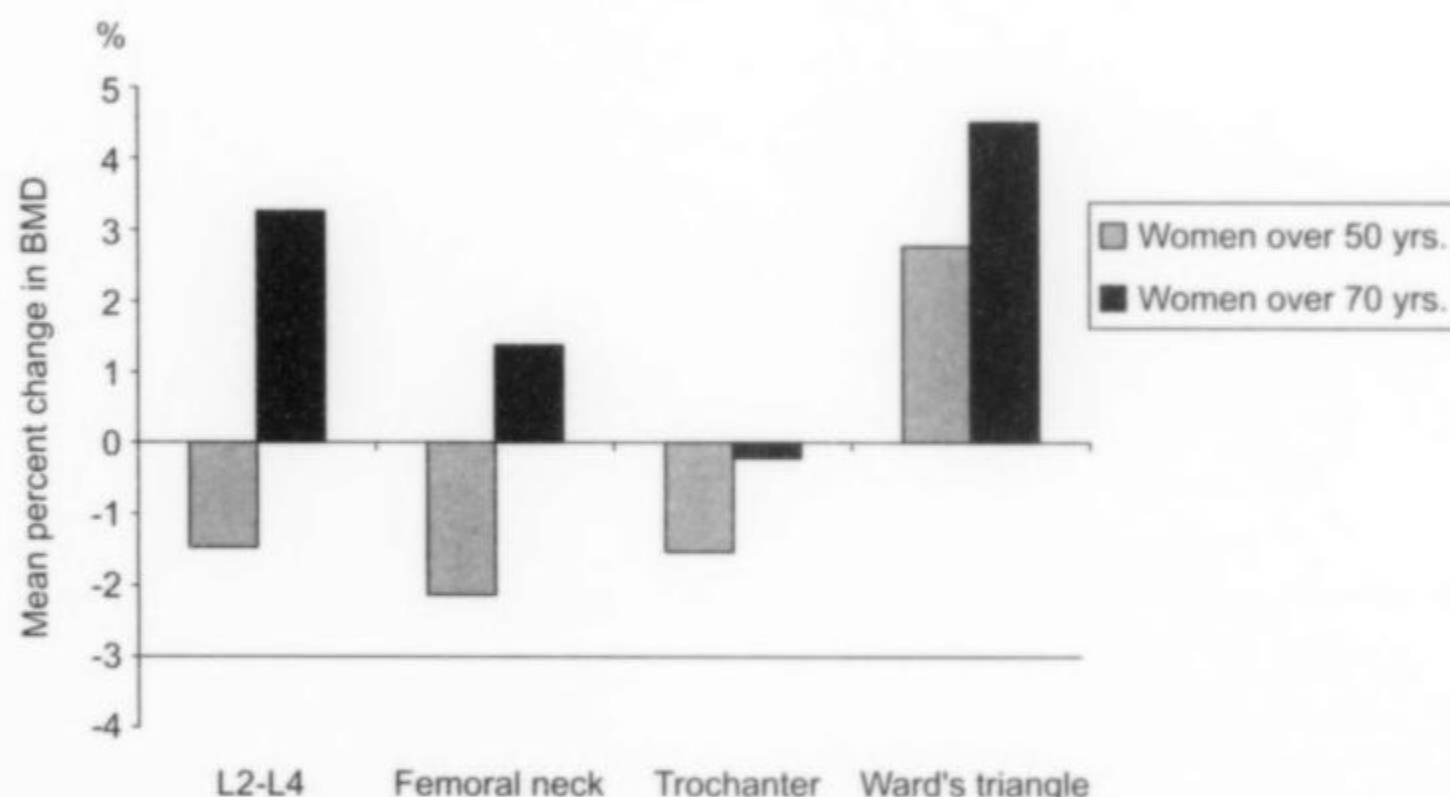


Fig. 1 Changes in bone density in postmenopausal women suffering from at least one vertebral fracture achieved after 1 year of treatment with Biomim H[®] (Masaryk *et al.* [38]).

serum phosphorus first decreased with a subsequent return to normal values. Urine calcium significantly increased after 3-6 months, subsequently decreasing again. Levels of osteocalcin, pyridinoline and deoxypyridinoline significantly decreased after eggshell powder treatment. The decrease in biochemical markers of osteoformation and osteoresorption, i.e. inhibition of bone turnover, after 6-month and 1-year administration of eggshell powder to postmenopausal women is also supported by a study by Stancíková *et al.* (39). Krišlo *et al.* (40) reported very good results with the treatment of vertebral pain syndromes of osteoporotic origin. In their patients, eggshell powder allowed long-term consumption of analgesics and other mainly symptomatic treatments to be reduced or discontinued. Notably, eggshell powder administration was associated with very few adverse effects apart from those common for calcium products, such as obstipation, meteorism, gastric discomfort, diarrhea and rare allergic reactions.

Interesting results were obtained by Hejčmanová *et al.* (41) in children and adolescents with various metabolic disorders. The effects of chicken eggshell powder on the prevention of osteopenia in children and adolescents with phenylketonuria was studied at the Child and Adolescent Clinics of the Charles University School of Medicine University Hospital in Prague, Czech Republic. Approximately 80-90% of common protein sources were removed from the diet of these children, thus making them eliminate the main sources of minerals which had to be replaced. A long-term study lasting 12 months, during which children and adolescents were given eggshell powder in doses of 550 mg of Ca per day, gave very good results. The concentrations of the osteoformation marker osteocalcin significantly increased, whereas those of the osteoresorption markers pyridinoline and deoxypyridinoline significantly decreased. The authors recommend eggshell powder as a reliable source of calcium in these patients.

Chicken eggshell powder is also useful in the treatment of Sudeck's syndrome. In addition to comprehensive and systematic physical therapy, it may successfully replace analgesics and trimepranol. Its administration in patients reduces swelling and skin temperature and increases the mobility of affected extremities (42). Šmondrk *et al.* (43) studied the effect of eggshell powder on symptoms of osteoarthritis in patients with coxarthrosis and gonarthrosis. They found pain-relieving properties and an increase in physical activity.

Friesland Coberko Dairy Foods is the largest producer of milk and dairy products in the Netherlands. This company has sponsored several projects focused on research into eggshells and it developed a product intended to supplement calcium in milk on the basis of chicken eggshell powder produced by Biomin a.s. Another Dutch company, Gull Nutraceutical B.V., continues research and development into new products. Several concepts of food supplements for children, adults and the elderly have been successfully developed so far. The key people in the research and promotion of eggshell powder in the Netherlands and other Western European countries have been I. Pakan and A. Schaafsma.

A pilot study on the effect of eggshell powder-enriched dairy products on BMD in patients with osteoporosis or osteopenia was also conducted by Schaafsma and Pakan (44-46). In nine postmenopausal women and one man they showed significant increases in the mean values of the lumbar spine, proximal femur and trochanter BMD bone mass density after a 4- to 8-month intervention; at the same time, a noticeable decrease was observed in some regions such as the lumbar spine and femoral neck in the control group of healthy women. Pain was also significantly reduced in patients with osteoporosis.

Schaafsma *et al.* (47) also evaluated the effect of supplementation of calcium (500 mg) and a standard mixture of vitamins and minerals on the BMD of the

lumbar spine and femoral neck in healthy women who had been menopausal for at least 5 years. Eighty-five women were divided into three groups: in group A, calcium was supplemented in the form of chicken eggshell powder and in group B calcium was supplemented in the form of pure calcium carbonate. The placebo group (group C) was given skimmed milk powder containing 25 mg of calcium and trace amounts of minerals and vitamins. The duration of the blinded, placebo-controlled study was 12 months. In addition to BMD, the authors also studied biochemical markers of bone metabolism. The results after 12 months showed that in women who were receiving calcium in the form of eggshell powder, femoral neck and Ward's triangle BMD values significantly increased compared with baseline values. The increase was not significant in the remaining groups. After 12 months, the difference in femoral neck BMD between group A and the placebo group C was in favor of eggshell powder. Among the biochemical indicators, a decrease in urine deoxypyridinoline was observed in group A when compared with group C. The results were thus more favorable for the women receiving calcium in the form of the eggshell powder.

Schaafsma et al. (4) performed a similar study in postmenopausal women with osteopenia and osteoporosis. In an open study, they studied 24 women with osteopenia and 26 with osteoporosis of the lumbar spine. The duration of study was 12 months. In this study, eggshell powder also provided better results than calcium carbonate: BMD values of trochanter and Ward's triangle increased only in the group of women taking eggshell powder.

Clinical and experimental studies show that eggshell powder has positive effects on bone and cartilage and that it is a suitable source of calcium in the prevention and treatment of osteoporosis.

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